Standard Operating Procedure (SOP)

Procurement and Installation of GPS based VTS Device by Vehicle Owners from Suggested Vendors

a) Objective

This SOP defines the process for vehicle owners to procure and install AIS-140 compliant Vehicle Tracking System (VTS) devices from Suggested vendors to ensure compliance with Directorate of Mines and Geology, Rajasthan guidelines.

b) Scope

Applicable to:

- Vehicle Owners / Fleet Operators mandated to install VTS devices.
- Suggested Vendors authorized by Directorate of Mines and Geology, Rajasthan
- Installation Centers approved by vendors and verified by Directorate of Mines and Geology, Rajasthan

c) Roles and Responsibilities

Stakeholder	Responsibilities
Vehicle Owner	Select vendor, make payment, present vehicle for installation, ensure compliance.
Suggested Vendor	Supply certified device, install, configure, and integrate with the monitoring system.
Directorate of Mines and Geology	Publish vendor list, verify compliance post-installation, maintain central VTS database.

d) Process Workflow

Steps	Activities
Step-1	Access Suggested Vendor List *Vehicle owner visits the Rajasthan-Mines portal and downloads/view list.*
Step-2	Vendor Selection and Order Placement *Contact chosen vendor, confirm order, and make payment*.
Step-3	Device Allocation and Appointment Scheduling *Vendor assigns device serial number and installation slot*
Step-4	Installation and Configuration *Install device, configure SIM and platform integration.*
Step-5	Data Integration and Compliance Update *Vendor updates details in Raj-Mines' VTS portal and generates Installation Certificate.*
Step-6	Completion and Documentation *Vehicle owner receives installation certificate, warranty, user manual.*

e) Service Level Agreement (SLA)

- Device Installation Time : Within 3 working days of order confirmation.
- Device Replacement : Within 48 hours in case of failure under warranty.
- Technical Support : 24x7 helpline for connectivity and functional issues.

f) Compliance and Penalty

- Non-installation within stipulated time may attract penalty as per Directorate of Mines and Geology.
- Vendor must maintain device uptime ≥ 99% and ensure data transmission to the central system.

- Failure to comply may result in vendor delisting.

g) Documentation Checklist

- Proof of Payment.
- AIS-140 Compliance Certificate for the device.
- Installation Certificate issued by the vendor.
- Owner's KYC (if required by regulation).

h) Annexure

Annexure 1: Installation Checklist
Annexure 2: Payment Receipt Template
Annexure 3: Installation Certificate Template
Annexure 4: VTS Device Technical Specifications
Annexure 5: AMC and Support Details

Annexure 1
Installation Checklist

Vehicle Owner Name	
Vehicle Registration Number	
Device IMEI Number	
Date of Installation	
GPS Signal Tested (Yes/No)	
SIM Activated (Yes/No)	

Annexure 2
Payment Receipt

Receipt No : ____________________________

Date : ________________________

Received from : ____

Amount Paid : ₹ ________________________________

Mode of Payment : _____________________________________

For VTS Device IMEI No : ______________________________

Vendor Name & Signature : ______________________________

Annexure 3
Installation Certificate

This is to certify that the VTS Device with the following details has been successfully installed:

Vehicle Registration Number : ____________________________________

Device IMEI Number : ____________________________________

Date of Installation : ____________________________________

Installed By : ____________________________________

Vendor Authorized Signatory : ____________________________________

Seal & Stamp

Annexure 4

AIS 140 GPS based VTS Device Technical Specifications

1) **Device Specifications**

a) **GNSS (GPS) Module**

- Position Accuracy: ≤ 2.5 meters CEP
- Cold Start: < 120 seconds
- Hot Start: < 10 seconds
- Antenna: Internal or External GNSS Antenna

b) **LTE Connectivity**

- Network Support: 4G fallback to 2G
- SIM Type: Embedded SIM
- GPRS Class: Class 10 or higher
- Data Transmission: TCP/IP, SMS fallback

c) **Processor & Memory**

- Flash Memory: Minimum 40000 Location Record or 4 hours of data when network is unavailable.
- Auto-upload on network restoration (store and forward mechanism).

d) **Digital / Analog Inputs & Communication Interface**

- Minimum 4 Digital Inputs (e.g. ignition, door, panic button)
- 2 Digital Outputs
- 2 Analog Inputs.
- 1 serial communication (e.g. RS232) for interfacing external systems.

e) **Power Supply**

- Operating Voltage: Device shall be designed to operate between 8VDC and 32VDC using vehicle battery input voltage range 12 /24Volts.
- Backup Battery: Device shall have an internal back-up battery to support 4 hours of normal operations (to be tested for positional record transmission at a frequency of 60 sec).
- Protections: Reverse Polarity Protection and Overvoltage Protection and Resilience (Surges/Spikes)
- Operating Temperature: −10°C to +70°C

f) **Enclosure & Indicators**

- Enclosure: Rugged enclosure with minimum IP65 protection.
- Tamper-proof design with sealable mounting and provision for secure installation inside the vehicle.

2) **Software & Functionality**

a) **Data Logging & Storage**

- **During Vehicle Movement (Operation):** The fixed frequency shall be user configurable, but the **minimum frequency shall be 5 seconds** during vehicle operation.
- **When Idle (Sleep/IGN OFF):** The fixed frequency shall be user configurable, but the data transmission rate shall be **not less than 10 minutes** in sleep/IGN OFF mode.

b) **Security & Compliance**

- Secure Protocol: Communication from the device to the backend must occur on a Secure channel over TCPIP protocol, preferably using socket-based connections that are authenticated and identifiable to prevent spoofing on IMEI/Unique ID.

c) **Geo-Fencing & Alerts**

- Geo-Fencing: Configurable multiple geofences
- Alert Types: As per AIS140 standard.

d) **Server Communication & Software Integration**

- Support AIS 140 protocol for real-time data transmission

3) **Certifications & Legal Compliance in India**

- AIS 140 Certification: By ICAT / ARAI / BIS / Other Govt. Labs
- IP Rating: Minimum IP65
- Device shall comply with AIS 140 environmental and EMC/EMI standards.
- Warranty: As per AIS140 standard.

4) **Accessories & Installation**

- Accessories: All required cables, mounting brackets, and panic buttons must be provided.
- Installation: As per AIS140 standard.

Annexure 5
AMC & Support Details

Vendor shall provide the following AMC and support services:

1) AMC Duration: ________ years (From ______________ To __________________)

2) Support Availability: 24x7 / Business Hours

3) Response Time: _____ hours

4) Contact Details for Support:

   Phone: ________________

   Email: ________________

5) Escalation Matrix:

Level 1: ____

Level 2: ____

Level 3: ____

AIS Standard Document Format

Table of Contents

1. Introduction

1.1 Purpose

The purpose of this document is to outline the TCP packet format used to communicate with the server for the vehicle tracking system.

1.2 Scope

This document encompasses the functional and non-functional requirements of the software system to be developed. It defines the features, constraints, and specifications that the software must adhere to in order to comply with the defined packet format.

2. Packet Formats

2.1. Location packet:

This packet provides details for the vehicle location.

Start Character, Vendor_ID, Firmware Version, Packet Type, Alert Id, Packet status, IMEI Number, Vehicle Reg. Number, GPS Fix, Date, Time, Latitude, Latitude Direction, Longitude, Longitude Direction, Speed, Heading angle, No of Satellite, Altitude, PDOP, HDOP, Network Operator, Ignition status, Main Power Status, Main Input Voltage, Internal Battery Voltage, Emergency Status, Tamper Status, GSM Signal Strength, Mobile Country Code, Mobile Network Code, Location Area Code, Cell ID, NMR, Digital Input Status, Digital Output Status, Frame Number, Checksum, End Character
e.g.
$1,VENDOR_ID,1.0.0,NR,1,L,867018065013637,XXXXXXXXXX,1,28032024,094604,17.038626,N,74.60238 6,E,0.23,192.29,19,591.70,1.14,0.80,BSNL3G,1,1,23.98,3.94,0,C,22,404,66,23B7,1FA0457,10782|-614|162 10782|-614|162 10782|-614|162 10782|-614|162,0000,00,000150,14150,*

The following table lists all the parameters from the location packet along with the description and sample data for each parameter.

Field	Description	Sample Data
Start Character	The header of the packet	$
Packet Header	Packet identifier	1,
Vendor_ID	Vendor Identification header	VENDOR_ID,
Firmware Version	Version details of the Firmware used in EX.1.0.0	1.0.0,
Packet Type	Specify the packet type	NR,
	NR = Normal	
	EA = Emergency Alert	
	TA = Tamper Alert (Optional)	
	HP = Health Packet	
	IN = Ignition On	
	IF = Ignition Off	
	BD = Vehicle Battery Disconnect	
	BR = Vehicle Battery Reconnect	

Field	Description	Sample Data
	BL = Internal Battery Low	
Alert Id	Please refer “Alerts packet Types” section for more information	1,
Packet Status	L=Live or H= History	L,
IMEI	Identified of the sending unit. 15 digit standard unique IMEI no.	867018065013637,
Vehicle Reg. No	Mapped vehicle registration number	XXXXXXXXXX,
GPS Fix	1 = GPS fix OR 0 = GPS invalid	1,
Date	Date value as per GPS date time per GPS date time (DDMMYYYY)	28032024,
Time	Time value as per GPS date time in UTC format (hhmmss)	094604,
Latitude	Latitude value in decimal degrees (not less than 6 places)	17.038626,
Latitude Dir	Latitude Direction. Example	N,
	N=North, S= South	
Longitude	Longitude value in decimal degrees (not less than 6 places).	74.602386,

Field	Description	Sample Data
Longitude Dir	Longitude Direction. E=East, W= West	E,
Speed	Speed of Vehicle as Calculated by GPS module in VLT. (in km/hrs.) (Upto One Decimal Value)	0.23,
Heading	Course over ground in degrees	192.29,
No of Satellites	Number of satellites available for fix	19,
Altitude	Altitude of the device in meters	591.70,
PDOP	Positional dilution of precision	1.14,
HDOP	Horizontal dilution of precision	0.80,
Network Operator Name	Name of Network Operator	BSNL 3G,
Ignition	1= Ignition On , 0 = Ignition Off	1,
Main Power Status	1= Vehicle battery connected 0=Vehicle battery disconnected	1,
Main Input Voltage	Indicator showing source voltage in Volts.(Upto One Decimal Value)	23.98,

Field	Description	Sample Data
Internal Battery Voltage	Indicator for level of battery charge remaining. (Upto One Decimal Value)	3.94,
Emergency Status	1= On , 0 = Off	0,
Tamper Alert (Optional)	C = Cover Closed, O = Cover Open	C,
GSM Signal Strength	Value Ranging from 0 – 31	22,
MCC	Mobile Country Code	404,
MNC	Mobile Network Code	66,
LAC	Location Area Code	23B7,
Cell ID	GSM Cell ID	1FA0457
NMR (Network Measurement Report)	Neighbouring 4 cell ID along with their LAC & signal strength	10782|-614|162
Neighbouring Cell ID		10782|-614|162
		10782|-614|162
		10782|-614|162,
Digital Input Status	4 external digital input status (Status of Input 1 to Input 3 (1=Off; 0=On))	0000,
Digital Output Status	2 external digital output status (1=On,0=Off)	00,
Frame Number	Sequence Number of the messages (000001 to 999999)	000150,
Checksum	Ensures No error in transmission (optimal)	14150,
End Character	Indicated End of the frame	*

2.2. Alert packet:

The Alert Packet is the same as location packet. Only Packet type and alert Id get changed according to the table given below.

Start Character, Vendor_ID, Firmware Version, Packet Type, Alert Id, Packet status, IMEI Number, Vehicle Reg. Number, GPS Fix, Date, Time, Latitude, Latitude Direction, Longitude, Longitude Direction, Speed, Heading angle, No of Satellite, Altitude, PDOP, HDOP, Network Operator, Ignition status, Main Power Status, Main Input Voltage, Internal Battery Voltage, Emergency Status, Tamper Status, GSM Signal Strength, Mobile Country Code, Mobile Network Code, Location Area Code, Cell ID, NMR, Digital Input Status, Digital Output Status, Frame Number, Checksum, End Character

e.g.

$1,VENDOR_ID,1.0.0,TA,9,L,868999039756527,MH12X2800,0,12032019,091034,73.50323,S,13.32442,E,0.0 00000,0.000000,0,0.000000,0.000000,0.000000,airtel,1,1,12.0,4.1,0,O,13,404,90,0c63,0529,4972|0c63|-94 5715|0c63|-95 052a|0c63|-97 4971|0c63|-103,1111,00,000008,242,*

The following table lists all the parameters from the alert packet along with the description and sample data for each parameter.

Name	Packet Type	Alert Id	Packet status	Remark
Normal packet	NR	1	L	Default message coming from each device
History packet	NR	2	H	Would be sent, if GPRS is not available at the time of sending the message in protocol format Zero, BLANK, NIL, etc
Disconnect main battery	BD	3	L/H	If device is disconnected from vehicle battery and running on its internal battery
Mains Battery connected	BR	6	L/H	I Indicates that device is connected back to main battery
Internal battery low	BL	4	L/H	If device internal battery has fallen below a defined threshold
Internal battery charged	BL	5	L/H	Indicates that device internal battery is charged again
Alert ignition ON	IN	7	L/H	Indicates that Vehicle's Ignition is switched ON

Name	Packet Type	Alert Id	Packet status	Remark
Alert ignition OFF	IF	8	L/H	Indicates that Vehicle's Ignition is switched OFF
Box open	TA	9	L/H	Optional message would be generated indicating GPS box opened
Box close	TA	9	L/H	Optional message would be generated indicating GPS box closed
Wire cut	TA	16	L/H	Emergency wire disconnected
Wire Reconnect	TA	16	L/H	Emergency wire reconnected
Emergency ON	EA	10	L/H	When any of the emergency button is pressed
Emergency OFF	EA	11	L/H	When emergency state of vehicle is removed
Harsh breaking	HB	13	L/H	Alert indicating for harsh braking
Harsh Acceleration	HA	14	L/H	Alert indicating for harsh acceleration
Rash Turning	HT/RT	15	L/H	Alert indicating for Rash turning
Over speed	OS	17	L/H	Indicates vehicle speed above threshold (Optional)
Alert Over the air parameter change	NR	12	L/H	When any parameter is changed over the air. Shall include the name of parameter changed and source of command

2.3. Login Message:

This section details the login message packet and its parameters.

Start Character, Vehicle Registration Number, IMEI, Firmware Version, Protocol Version, Latitude, Latitude Direction, Longitude, Longitude Direction, End character

e.g.
$1,XXXXXXXXXX,867018065013637,1.0.0,PRTCL_1.0,17.038708,N,74.602516,E,*

The following table lists all the parameters from the login packet along with the description and sample data for each parameter.

Field	Description	Sample Data
Start Character	The header of the packet	$
Packet Header	packet identifier	1,
Vehicle Registration Number	Mapped vehicle registration number	XXXXXXXXXX,
IMEI	Identified of the sending unit. 15 digit standard unique IMEI no.	867018065013637,
Firmware Version	Version details of the Firmware used in EX.1.0.0	1.0.0,
Protocol Version	Protocol version used for communication with server	PRTCL_1.0,
Latitude	Latitude value in decimal degrees (not less than 6 places)	17.038708,
Latitude Direction	Latitude Direction. Example N=North, S= South	N,
Longitude	Longitude value in decimal degrees (not less than 6 places).	74.602516,
Longitude Direction	Longitude Direction. E=East, W= West	E,
End Character	Indicated End of the frame	*

2.4. Health Message:

Below are the details of the health message packet.

Start Character, Vendor_ID, Firmware Version, Packet Type, IMEI, Battery Percentage, Low Battery Threshold Value, Memory Percentage, Data Update Rate When Ignition ON, Data Update Rate When Ignition OFF, Digital I/O Status, Analog 1 I/O Status, Analog 2 I/O Status, Analog 3 I/O Status, End Character

e.g.
$1,VENDOR_ID,513716301109_0D,HP,867018065013637,89,35,70,10,3600,0000,0.00,0.00,*

The following table lists all the parameters from the health packet along with the description and sample data for each parameter.

Field	Description	Sample Data
Start Character	The header of the packet	$1,
Vendor_ID	Vendor Identification header	VENDOR_ID,
Firmware Version	Version details of the Firmware used in EX.1.0.0	1.0.0,
Packet Type		HP,
IMEI	Identified of the sending unit. 15 digit standard unique IMEI no.	867018065013637,
Battery percentage	Indicates the internal battery charge percentage	89,
Low battery threshold value	Indicates value on which low battery alert generated in percentage	35,
Memory percentage	Indicates flash memory percentage used	70,
Data update rate when ignition ON i.e. NPIN on IGN ON	Indicates NR Packet frequency on ignition ON	10,

Field	Description	Sample Data
Auto wakeup timer	Indicates Wakeup time interval.	3600,
Digital I/o status	Inputs connected to the device.	0000,
Analog 1 I/o status	Analog 1 input status	0.00,
Analog 2 I/o status	Analog 2 input status	0.00,
End character	*	*

2.5. Emergency Packet For Second Server:

This section details the emergency packet data.

Start Character, Header, Packet type, IMEI, Packet Status, Date, Time, GPS Validity, Latitude, Latitude Direction, Longitude, Longitude Direction, Altitude, Speed, Distance, GPS Provider, Vehicle Registration Number, Reply Number, End Character, Checksum

e.g.

$EPB,EMR,867018065013637,NM,29032024
054225,A,17.038746,N,74.602303,E,578.54,0.39,0.00,G,XXXXXXXXXX,9175680139,*6225

The following table lists all the parameters from the emergency packet along with the description and sample data for each parameter.

Field	Description	Sample Data
Start Character	The header of the packet	$
Packet Header	The unique identifier for emergency messages from VLT	EPB,
Packet Type	Message Types supported. Emergency Message (EMR) or Stop Message (SEM)	EMR,
IMEI Number	Unique ID of the Vehicle	867018065013637,

Field	Description	Sample Data
	(IMEI Number)	
Packet Status	NM – Normal Packet, SP – Stored Packet	NM,
Date and time	Date and time of location the location obtained from the data in DDMMYYYY hhmmss format	29032024 054225,
GPS Validity	V/A – Valid, N – Invalid	A,
Latitude	Latitude in decimal degrees - dd.mmmmmm format	17.038746,
Latitude Direction	N – North, S – South	N,
Longitude	Longitude in decimal degrees - dd.mmmmmm format	74.602303,
Longitude Direction	E – East W – West	E ,
Altitude	Altitude in meters (above sea level)	578.54,
Speed	Speed of Vehicle as Calculated by GPS module in VLT. (in km/hr)	0.39,
Distance	Distance calculated from previous GPS data	0.00,
Provider	G - Fine GPS N - Coarse GPS or data from the network	G,
Vehicle Reg. No	Registration Number of the Vehicle	XXXXXXXXXX,

Field	Description	Sample Data
Reply Number	The mobile number to which Test response needs to be sent. (Emergency Mobile No. as specified by MHA/MoRTH/States.)	9175680139,
End Character	*	*
Check sum	Ensure no error in transmission	6225

2.6 Activation message and Health Check Message Protocol

The protocols for activation message and health check message are given below.

Device shall send the activation and health check messages on request as specified below directly to the backend system (i.e. backend Command and Control Centre set up/ authorized by State/UT or a Common Layer system providing interface to VLT device manufacturers' backend applications).

A. Activation SMS Format from Backend System to Device
B. For completion of the installation process, the VLT device shall undergo Activation process as per below:

• Activation Message Request Format from the Backend System to the Device

(Through SMS): ACTV, Random Code, Reply SMS Gateway no.

• Activation Message Reply Format from Device to the Backend System

(Through SMS) as per below:

The following table lists all the parameters from the activation message packet along with the description and sample data for each parameter.

Field Name	Characters	Activation Example
Header	5	ACTVR
Separator	1	,
Random code	6	123456

Field Name	**Characters**	**Activation Example**
Separator	1	,
Vendor_ID	4	VENDOR_ID
Separator	1	,
Firmware version	6	1.0.0
Separator	1	,
IMEI	15	867018065013637
Separator	1	,
Alert ID	2	17
Separator	1	,
Latitude	12	17.038649
Separator	1	,
Latitude direction	1	N
Separator	1	,
Longitude	12	74.602493
Separator	1	,
Longitude direction	1	E
Separator	1	,
GPS fix	1	1
Separator	1	,

Field Name	Characters	Activation Example
Date and Time	15	29032024 112534
Separator	1	,
Heading Angle	6	183.58
Separator	1	,
Speed	4	0.09
Separator	1	,
GSM Strength	2	21
Separator	1	,
Country Code (MCC)	3	404
Separator	1	,
Network Code (MNC)	4	90
Separator	1	,
LAC	4	1368
Separator	1	,
Main Power Status	1	1
Separator	1	,
IGN Status	1	1
Separator	1	,

Field Name	Characters	Activation Example
Mains Battery Vtg	4	24.06
Separator	1	,
Frame Number	6	0
Separator	1	,
Vehicle mode	2	0

B. Health Check Random Messages from Backend System to Device Frequency: Twice Daily

(Recommended), Health Check Message Request Format from the Backend System to the Device (Through SMS): HCHK, Random Generated ID, Reply SMS Gateway no.

Health Check Message Reply Format from Device to Backend System (Through SMS):

The following table lists all the parameters from the health check message packet along with the description and sample data for each parameter.

Field Name	Characters	Activation Example
Header	5	HCHKR
Separator	1	,
Random code	6	123456
Separator	1	,
Vendor_ID	4	VENDOR_ID
Separator	1	,
Firmware version	6	5.2.8_TST17
Separator	1	,
IMEI	15	867018065013637
Separator	1	,

Field Name	Characters	Activation Example
Alert ID	2	17
Separator	1	,
Latitude	12	17.038778
Separator	1	,
Latitude direction	1	N
Separator	1	,
Longitude	12	74.602448
Separator	1	,
Longitude direction	1	E
Separator	1	,
GPS fix	1	1
Separator	1	,
Date and Time	15	29032024 113452
Separator	1	,
Heading Angle	6	329.18
Separator	1	,
Speed	4	0.4
Separator	1	,
GSM Strength	2	21
Separator	1	,

Field Name	Characters	Activation Example
Country Code (MCC)	3	404
Separator	1	,
Network Code (MNC)	4	90
Separator	1	,
LAC	4	1368
Separator	1	,
Main Power Status	1	1
Separator	1	,
IGN Status	1	1
Separator	1	,
Mains Battery Vtg	4	24.08
Separator	1	,
Frame Number	6	1
Separator	1	,
Vehicle mode	2	0

2.7 Configuration of Device (OTA)

The device shall support at least the below parameters to be configurable over the air (through SMS and Cellular network). The updation/ configuration shall be allowed only over an 'authenticated' channel.

1. Setting/ Change of the Primary or Secondary IP and port number
2. Setting/ Change of the APN
3. Set configuration parameters like sleep time, over speed limit, harsh braking, harsh acceleration, rash turning threshold limits etc.

4. Emergency control SMS Centre Number(s)
5. Configuring the vehicle registration number
6. Configuring the frequency of data transmission in normal / Ignition state / OFF state sleep mode/ Emergency state, etc.
7. Configuring the time duration for Emergency state
8. Capability to reset the device
9. Command to get the IMEI of the device Configurable commands must involve the following features:
9.1. SET: For setting the parameters.
9.2. GET: For enquiring regarding the parameters such as mobile number, GSM strength, vehicle number and other important parameters.
9.3. CLR: For clearing certain commands, alarms, alerts etc. except emergency alert After each SET, GET, CLR command the device should send alert to Backend Control Centre.

e.g. OTA commands

The following table includes example OTA commands along with the format and sample for each. All OTA list will be provided as per request by backend team for integration.

Command Description	**Format**	**Sample data**
Setting/ Change of the Primary or Secondary IP and port number	*SET,CHTP,IPAddress,Port#	*SET,CHTP,1.186.231.101,5003#
	*SET,CIP1,IPAddress,Port#	*SET,CIP1,1.186.231.101,5001#
SOS Acknowledgement	*SET,SACK,VAL#	*SET,SACK,1#
Emergency Mobile number	*SET,CMNU,Mobile Number#	*SET,CMNU,9120045868#
Access Point Name (APN)	*SET,CGPR,APN#	*SET,CGPR,"sensem2m2",""#
Vehicle Registration number	*SET,VREGN,Vehicle No.#	*SET,VREGN,AP16CA8136#
GET Configuration	*GET,CHTP#	
	*GET,CMNU#	
Clear configuration	*CLR,CHTP#	